

# DS 5 LIBRARY PROGRAMS

## DSA0404-FDS

Name: K. L. Khaviya varshini

Reg. No.: 192472381

### GROUPING:

```
GROUPING.py - C:/Users/Khaviya KL/OneDrive/Desktop/DS 5 LIBRARY PROGRAMS/GROUPING.py (3.13.5)
File Edit Format Run Options Window Help
import pandas as pd
data = {
    'Name': ['Arun', 'Bala', 'Chitra', 'Divya', 'Ezhi', 'Fathima'],
    'Department': ['CSE', 'ECE', 'CSE', 'ECE', 'CSE', 'ECE'],
    'Marks': [85, 78, 92, 88, 75, 95]
}

df = pd.DataFrame(data)
print("Average Marks by Department:")
grouped_df = df.groupby('Department')['Marks'].mean()
print(grouped_df)
```

```
IDLE Shell 3.13.5
File Edit Shell Debug Options Window Help
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: C:/Users/Khaviya KL/OneDrive/Desktop/DS 5 LIBRARY PROGRAMS/GROUPING.py =====
Original Data:
   Name Department Marks
0  Arun         CSE    85
1  Bala         ECE    78
2 Chitra         CSE    92
3  Divya        ECE    88
4  Ezhi         CSE    75
5 Fathima        ECE    95

Sorted by Marks (Ascending):
   Name Department Marks
4  Ezhi         CSE    75
1  Bala         ECE    78
0  Arun         CSE    85
3  Divya        ECE    88
2 Chitra         CSE    92
5 Fathima        ECE    95

>>>
===== RESTART: C:/Users/Khaviya KL/OneDrive/Desktop/DS 5 LIBRARY PROGRAMS/GROUPING.py =====
Average Marks by Department:
Department
CSE    84.0
ECE    87.0
Name: Marks, dtype: float64

>>>
```

### SORTING:

```
SORTING.py - C:/Users/Khaviya KL/OneDrive/Desktop/DS 5 LIBRARY PROGRAMS/SORTING.py (3.13.5)
File Edit Format Run Options Window Help
import pandas as pd
data = {
    'Name': ['Arun', 'Bala', 'Chitra', 'Divya', 'Ezhi', 'Fathima'],
    'Department': ['CSE', 'ECE', 'CSE', 'ECE', 'CSE', 'ECE'],
    'Marks': [85, 78, 92, 88, 75, 95]
}

df = pd.DataFrame(data)
print("Original Data:")
print(df)

print("\nSorted by Marks (Ascending):")
sorted_df = df.sort_values(by='Marks', ascending=True)
print(sorted_df)
```

```
IDLE Shell 3.13.5
File Edit Shell Debug Options Window Help
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: C:/Users/Khaviya KL/OneDrive/Desktop/DS 5 LIBRARY PROGRAMS/SORTING.py =====
Original Data:
   Name Department Marks
0  Arun         CSE    85
1  Bala         ECE    78
2 Chitra         CSE    92
3  Divya        ECE    88
4  Ezhi         CSE    75
5 Fathima        ECE    95

Sorted by Marks (Ascending):
   Name Department Marks
4  Ezhi         CSE    75
1  Bala         ECE    78
0  Arun         CSE    85
3  Divya        ECE    88
2 Chitra         CSE    92
5 Fathima        ECE    95

>>>
```

## RANKING:

```
RANKING.py - C:/Users/Khavya KL/OneDrive/Desktop/DS 5 LIBRARY PROGRAMS/RANKING.py (3.13.5)
File Edit Format Run Options Window Help

import pandas as pd

data = {
    'Name': ['Arun', 'Bala', 'Chitra', 'Divya', 'Ezhil', 'Fathima'],
    'Department': ['CSE', 'ECE', 'CSE', 'ECE', 'CSE', 'ECE'],
    'Marks': [85, 78, 92, 88, 75, 95]
}

df = pd.DataFrame(data)

df['Rank'] = df['Marks'].rank(ascending=False, method='dense')

print("Students with Rank:")
print(df.sort_values(by='Rank'))
```

```
IDLE Shell 3.13.5
File Edit Shell Debug Options Window Help

Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.

>>> ===== RESTART: C:/Users/Khavya KL/OneDrive/Desktop/DS 5 LIBRARY PROGRAMS/SORTING.py =====
Original Data:
   Name Department Marks
0  Arun         CSE    85
1   Bala         ECE    78
2  Chitra         CSE    92
3  Divya         ECE    88
4  Ezhil         CSE    75
5 Fathima         ECE    95

Sorted by Marks (Ascending):
   Name Department Marks
4  Ezhil         CSE    75
1   Bala         ECE    78
0  Arun         CSE    85
3  Divya         ECE    88
2  Chitra         CSE    92
5 Fathima         ECE    95

>>> ===== RESTART: C:/Users/Khavya KL/OneDrive/Desktop/DS 5 LIBRARY PROGRAMS/GROUPING.py =====
Average Marks by Department:
Department
CSE    84.0
ECE    87.0
Name: Marks, dtype: float64

>>> ===== RESTART: C:/Users/Khavya KL/OneDrive/Desktop/DS 5 LIBRARY PROGRAMS/RANKING.py =====
Students with Rank:
   Name Department Marks Rank
5 Fathima         ECE    95   1.0
2  Chitra         CSE    92   2.0
3  Divya         ECE    88   3.0
0  Arun         CSE    85   4.0
1   Bala         ECE    78   5.0
4  Ezhil         CSE    75   6.0

>>>
```

Ln: 39 Col: 0

## PLOTTINGAVG:

```
PLOTTING AVG MARKS BY DEPT.py - C:/Users/Khavya KL/OneDrive/Desktop/DS 5 LIBRARY PROGRAMS/PLOTTING AVG MARKS BY DEPT.py (3.13.5)
File Edit Format Run Options Window Help

import pandas as pd
import matplotlib.pyplot as plt

data = {
    'Name': ['Arun', 'Bala', 'Chitra', 'Divya', 'Ezhil', 'Fathima'],
    'Department': ['CSE', 'ECE', 'CSE', 'ECE', 'CSE', 'ECE'],
    'Marks': [85, 78, 92, 88, 75, 95]
}

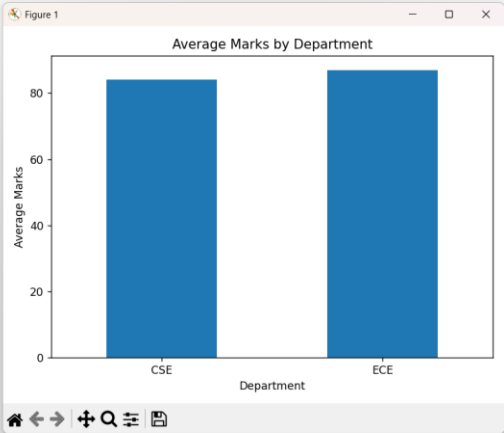
df = pd.DataFrame(data)

grouped_df = df.groupby('Department')['Marks'].mean()

grouped_df.plot(kind='bar')

plt.title("Average Marks by Department")
plt.xlabel("Department")
plt.ylabel("Average Marks")

plt.xticks(rotation=0)
plt.tight_layout()
plt.show()
```



| Department | Average Marks |
|------------|---------------|
| CSE        | 84.0          |
| ECE        | 87.0          |

Ln: 1 Col: 18

## PLOTTING MARKS:

